KINETICAI

Real-Time Pose Estimation & Form Coach

Complete Project Planning Document

Author	Navnit Amrutharaj
Version	1.0 — March 2026
Category	Browser AI · WebGPU · Health & Sports Tech
Stack	MoveNet + TF.js · MediaPipe · WebGPU · Gemini API · Vercel
Deployment Cost	$0 / month (fully free tier)
Target Completion	9 Weeks

1. Executive Summary

KineticAI is a **zero-install, browser-native real-time pose estimation and movement coaching platform** that runs entirely on the user's device using WebGPU acceleration — no server, no app download, no data ever leaving the camera. It uses **Google's MoveNet Lightning** (50+ FPS at 17 keypoints via TensorFlow.js) combined with **MediaPipe's BlazePose** (33 keypoints, 3D-aware) and the **Gemini 1.5 Flash free API** for AI-generated personalised correction feedback, all deployed on Vercel at zero cost.

KineticAI targets a **massive and underserved real-world problem**: the global physiotherapy and sports performance gap. The global physical therapy services market was valued at **USD 134.92 billion in 2024** and is growing at 3.6% CAGR, yet **a US physiotherapy session costs $30–$400** and insurance routinely caps covered visits. A 2026 clinical study published in PMC (n=146) proved that on-device AI pose estimation using MediaPipe achieves **95.8% accuracy vs. physiotherapist gold-standard assessment** (ICC = 0.91) — meaning the technology is

now clinically validated for home rehabilitation. KineticAI democratises this technology by delivering it in a browser, for free, to anyone with a camera.

🎯 **The Core Thesis**

Traditional physiotherapy and sports coaching have three hard constraints: cost ($30–$400 per session), access (must travel to a clinic), and scalability (one physiotherapist can see ~8 patients per day). KineticAI removes all three: it runs on any laptop or phone with a browser, costs $0 to use, and can serve an unlimited number of users simultaneously because all inference is client-side. The WebGPU architecture Navnit already proved with YOLOv11 — 15–30 FPS client-side — is directly applicable here at even higher frame rates (50–120 FPS with MoveNet Lightning).

2. Problem Statement & Real-World Context

2.1 The Access & Cost Crisis in Physiotherapy and Sports Coaching

Despite physiotherapy being the primary evidence-based treatment for musculoskeletal injuries, the majority of the world's population cannot consistently access it due to cost and geography. Key statistics:

- Global physical therapy market: USD 134.92B in 2024, projected USD 186.9B by 2033 — a vast industry with unmet demand
- US session cost: $30–$400 per session; insurance companies routinely limit covered visits, forcing patients to discontinue
- 90% of rehabilitation exercises are performed at home without direct supervision — the most dangerous phase of recovery
- AI-enhanced remote physiotherapy market growing at 24.9% CAGR — from USD 2.12B in 2025 to USD 5.16B by 2029
- Studies show clinicians achieve only 12° accuracy in angular measurements during visual assessment — human assessment is objectively imprecise
- MediaPipe/MoveNet pose estimation achieves 95.8% classification accuracy vs. physiotherapist gold-standard (PMC 2026 clinical trial, n=146, ICC=0.91)

2.2 The Technology Gap: Why Existing Solutions Fall Short

Solution	What it offers	Why it fails most users
In-person physiotherapy	Expert assessment, personalised plan	$30–$400/session; limited sessions; geography barrier; therapist shortage
Wearable sensors (IMU)	Accurate joint angle data	Hardware cost ($200–$2000); must be purchased, charged, calibrated per session
Commercial apps	AI movement coaching	Subscription $15–$30/month;

(Kemtai, Kaia)		closed platform; no customisation; data sent to cloud
Motion capture (Vicon, Kinect)	Gold-standard accuracy	Costs $10,000–$100,000; lab setting only; impossible for home use
Static pose apps (yoga apps)	Single-frame pose check	No real-time tracking; no joint angle measurement; no rep counting
KineticAI	Real-time joint angles, rep count, AI coaching	Free, no install, works in any browser, no data leaves device

2.3 The 2026 Technical Readiness Moment

📊 **Key Research Findings**

Three convergent developments make 2026 the right year: (1) A 2026 clinical trial (n=146, PMC) proved on-device AI pose estimation achieves 97.8% squat accuracy and 96.4% deadlift accuracy vs. physiotherapist — clinical validation was the missing proof. (2) MoveNet Lightning runs at 50+ FPS on CPU-only devices and 120 FPS on GPU — real-time feedback is now guaranteed on any modern browser. (3) WebGPU is now supported in Chrome, Edge, Safari, and Firefox — the hardware acceleration path is mainstream, not experimental. The combination makes a fully free, clinically-grade home physiotherapy assistant buildable today.

3. Project Vision, Name & Differentiators

3.1 Name: KineticAI

Kinetic — relating to motion and movement, the core domain of the project. Scientific, precise, and used in physiotherapy (kinetic chain, kinetic analysis). **AI** — clearly communicates the intelligence layer: it is not just a tracker but a coach that analyses, scores, and gives corrective feedback. Together, **KineticAI** is clean, technically precise, and immediately understood by both the health/sports audience and the tech audience reviewing a portfolio.

3.2 Vision Statement

KineticAI will be the **world's most accessible AI movement coach** — a browser-native tool that delivers real-time joint angle tracking, rep counting, form scoring, and AI-generated personalised feedback to anyone with a camera and an internet connection, completely free, with zero data privacy concerns because all inference runs on the user's own device.

3.3 Target Users & Use Cases

User	Primary Use Case	Key Need KineticAI Addresses
Physiotherapy patient (home)	Supervised home rehab exercises post-surgery/injury	Correct form without paying $400/session; 90% of rehab is unsupervised
Amateur athlete	Squat, deadlift, overhead press form correction	Prevent injury from incorrect technique without a personal trainer
Yoga / fitness enthusiast	Pose alignment during yoga, pilates, HIIT	Real-time cue-based correction without a video class subscription
Sports coach	Remote athlete form analysis	Shareable PDF session report to review with athletes asynchronously
Developer / researcher	Test WebGPU inference architecture	Open-source reference implementation of WebGPU + ONNX + pose estimation

3.4 Core Differentiators

Differentiator	What it means	Why others don't have it
100% client-side inference	MoveNet + WebGPU: all ML runs in browser, zero data sent to server	Competitors require cloud upload for analysis, creating latency and privacy risk
50–120 FPS real-time tracking	MoveNet Lightning on WebGPU: clinically validated sub-20ms inference	Wearables match this but cost $200+; cloud apps have 500ms+ round-trip latency
33-keypoint 3D-aware mode	MediaPipe BlazePose: includes wrist rotation, foot orientation, neck angle	MoveNet (17 keypoints) misses hand/foot/head orientation critical for physio
AI narrative feedback	Gemini 1.5 Flash free API generates personalised natural language coaching tips	Existing free tools give only colour-coded alerts — no explanation of why form is wrong
Shareable PDF session report	jsPDF: client-side PDF generation from session data — no server	No competitor offers a downloadable, shareable session record for free
Multi-exercise library	20+ exercises with golden-angle ranges per joint, configurable thresholds	Generic trackers don't know what 'good squat depth' is — KineticAI does
Zero-install, any device	Pure browser app on Vercel — works on any camera-equipped device	Wearables need pairing; native apps need download; KineticAI opens in a tab

4. System Architecture

4.1 High-Level Architecture

KineticAI is a **pure frontend application** — there is no backend server involved in the critical path. The architecture has four functional layers: the **Capture Layer** (camera → WebRTC MediaStream), the **Inference Layer** (MoveNet/BlazePose via TensorFlow.js on WebGPU), the **Analysis Layer** (joint angle engine, rep counter, form scorer), and the **Output Layer** (real-time overlay, Gemini feedback, PDF report).

```
ARCHITECTURE OVERVIEW — KineticAI (100% Client-Side)

  Camera (WebRTC)  →  HTMLVideoElement  →  requestAnimationFrame loop
         ↓
  TensorFlow.js  →  WebGPU backend  →  MoveNet Lightning / BlazePose
         ↓
  Pose keypoints (17 or 33 joints, x/y/z + confidence score per joint)
         ↓
  Joint Angle Engine  →  Calculates angles for all configured joint pairs
         ↓                (hip, knee, ankle, shoulder, elbow, spine, neck)
  ┌──────────────────────────────────────────────────────────────┐
  │ Analysis Layer (runs in Web Worker — non-blocking)           │
  │  ├─ Rep Counter: state machine on primary joint angle        │
  │  ├─ Form Scorer: per-joint deviation from golden range       │
  │  ├─ Phase Detector: eccentric / concentric / hold phase      │
  │  └─ Fatigue Detector: rep-over-rep angle degradation         │
  └──────────────────────────────────────────────────────────────┘
         ↓
  Canvas Overlay  →  Skeleton render + angle labels + colour-coded joints
         ↓
  Session Store (in-memory)  →  Keyframes per rep, score per rep, timestamp
         ↓
  On session end: Gemini 1.5 Flash API (free)  →  AI coaching feedback
         ↓
  jsPDF (client-side)  →  PDF session report  →  Download or share link
```

4.2 Model Strategy: MoveNet + BlazePose Dual Mode

KineticAI uses **two models** selectable by use case, not competing — they serve different needs:

Model	Keypoints	FPS (WebGPU)	Best For	Limitation
MoveNet	17 keypoints, 2D	50–120 FPS	Speed-critical	No wrist rotation,

Lightning			apps: squat, push-up, running — any exercise needing fast rep counting	no 3D depth, no foot orientation
MoveNet Thunder	17 keypoints, 2D	30–60 FPS	Slower exercises needing higher accuracy: yoga, deadlift, posture analysis	Slower; still no 3D
MediaPipe BlazePose	33 keypoints, 2D+3D	25–40 FPS	Full-body physiotherapy analysis, ACL injury risk, yoga alignment, neck/wrist assessment	Slower; higher compute; single person only

KineticAI's **model auto-selector** picks the right model based on the exercise chosen from the library. Fast compound exercises (squat, burpee, jump) use MoveNet Lightning. Slow, precision-critical exercises (deadlift, physio rehab, yoga) use BlazePose. The user can override the auto-selection.

> ⚡ **WebGPU Backend**
> TensorFlow.js now supports a WebGPU backend alongside WebGL. On devices with WebGPU support (Chrome 113+, Edge, Safari 18+), MoveNet Lightning runs at 50–120 FPS — comparable to native app performance. On older devices, TF.js automatically falls back to WebGL (30–50 FPS) or WASM (15–25 FPS). KineticAI detects the best backend at startup and shows the user the current FPS and backend in the status bar.

5. Joint Angle Engine

5.1 Architecture

The Joint Angle Engine is the analytical core of KineticAI. It takes raw keypoint coordinates from the pose model and computes **clinically meaningful joint angles** using vector mathematics. Every frame, it outputs a structured `PoseAnalysis` object with joint angles, phase classification, form scores, and rep state.

```
// Joint angle calculation — law of cosines on 3 keypoints
function calculateAngle(A, B, C) {
  // A = proximal joint (e.g. hip), B = vertex joint (e.g. knee), C =
distal (e.g. ankle)
  const BA = { x: A.x - B.x, y: A.y - B.y };
```

```
  const BC = { x: C.x - B.x, y: C.y - B.y };
  const dot = BA.x * BC.x + BA.y * BC.y;
  const magBA = Math.sqrt(BA.x**2 + BA.y**2);
  const magBC = Math.sqrt(BC.x**2 + BC.y**2);
  return Math.acos(dot / (magBA * magBC)) * (180 / Math.PI);
}

// Example: Knee angle during squat
const kneeAngle = calculateAngle(
  keypoints[LEFT_HIP],   // A
  keypoints[LEFT_KNEE],  // B (vertex)
  keypoints[LEFT_ANKLE]  // C
);
// 170–180° = standing, 90° = parallel squat, <70° = below parallel
```

5.2 Joint Pairs Tracked Per Exercise Category

Exercise Category	Primary Joints	Key Angles Measured
Lower body strength (squat, lunge, deadlift)	Hip, Knee, Ankle	Knee flexion, hip hinge, forward lean angle
Upper body strength (push-up, press, row)	Shoulder, Elbow, Wrist	Elbow flexion, shoulder abduction, wrist angle
Core & posterior chain (plank, bridge, deadlift)	Hip, Spine, Shoulder	Hip extension, spinal neutral, trunk angle
Physiotherapy rehab (shoulder, knee, hip)	Specific to exercise	Range of motion percentage, symmetry L vs R
Yoga & mobility	All joints	Multi-joint alignment, depth achievement
Running gait (video upload mode)	Hip, Knee, Ankle	Cadence, stride angle, overstriding

6. Rep Counter & Form Scorer

6.1 Rep Counter — State Machine Architecture

The rep counter uses a **finite state machine** driven by the primary joint angle of an exercise. For a squat, the primary angle is knee flexion. The state machine has four states: `STANDING`,

`DESCENDING`, `BOTTOM`, `ASCENDING`. A valid rep requires passing through all four states in order, with configurable angle thresholds per exercise.

```
// Rep counter state machine — squat example
const SQUAT_CONFIG = {
  primaryJoint: 'LEFT_KNEE',
  standingAngle: { min: 165, max: 180 },   // degrees
  bottomAngle:   { min: 60,  max: 100 },    // parallel = ~90°
  repCompletes:  'on_ascend_past_standing',
  minRepDuration: 1200,   // ms — prevents counting bounced reps
};

// State transitions
STANDING    → DESCENDING : knee angle drops below 150°
DESCENDING  → BOTTOM     : knee angle drops below bottomAngle.max
BOTTOM      → ASCENDING  : knee angle begins increasing (derivative > 0)
ASCENDING   → STANDING   : knee angle exceeds standingAngle.min → REP
COUNTED
```

6.2 Form Scorer — Per-Rep Quality Score

Every rep receives a **Form Score (0–100)** based on how well the user's joint angles matched the golden range for each exercise. The score is a weighted average of all monitored joints:

Score Component	Weight	What it measures
Depth achievement	30%	Did the primary joint reach the target angle range? (e.g. knee to 90° for squat)
Alignment deviation	25%	Was the spine neutral? Did knees track over toes? Was the neck in line?
Symmetry (L vs R)	20%	Were both sides of the body within 10° of each other throughout the rep?
Control (speed)	15%	Was the eccentric phase controlled (> 1.5 seconds)? Rushed reps score lower.
Full range of motion	10%	Did the joint return to starting position fully? Incomplete reps score lower.

6.3 Fatigue Detection

KineticAI tracks **rep-over-rep degradation** to estimate fatigue and warn the user before injury risk increases:

- Depth fade: if knee angle at bottom position increases by > 10° over 3 consecutive reps, fatigue warning fires
- Symmetry collapse: if L/R asymmetry exceeds 15° for 2 consecutive reps, compensation warning fires
- Speed collapse: if rep duration drops > 30% below the user's baseline rep speed, pacing warning fires
- Form trend: if the rolling 5-rep average form score drops below 70, session rest recommendation fires

7. Exercise Library (20+ Exercises)

7.1 Library Structure

Every exercise in KineticAI is defined by a **JSON configuration object** that specifies the model to use, the primary and secondary joints to track, the golden angle ranges, the form score weights, and the coaching cues. This makes the library infinitely extensible — adding a new exercise requires no code changes, only a new JSON entry.

```
// Exercise config example: Romanian Deadlift
{
  id: 'romanian-deadlift',
  name: 'Romanian Deadlift',
  model: 'blazepose',         // needs hip hinge angle (3D-aware)
  category: 'lower-body-strength',
  primaryJoint: { vertex: 'HIP', proximal: 'SHOULDER', distal: 'KNEE' },
  goldenRanges: {
    hipAngle:       { bottom: { min: 45, max: 80 }, top: { min: 160, max:
180 } },
    kneeAngle:      { bottom: { min: 155, max: 175 } },   // slight bend
only
    spinalDeviation: { max: 10 },                          // degrees from
neutral
  },
  coachingCues: {
    tooUpright:   'Push your hips back further — hinge, don\u0027t squat',
    kneesBent:    'Keep a soft knee throughout — avoid excessive bending',
    backRounded:  'Brace your core and maintain a flat back throughout',
    goodForm:     'Perfect hip hinge pattern — neutral spine maintained',
  },
  targetReps: 10,
  restSeconds: 90,
}
```

7.2 Exercise Library Contents

Category	Exercises Included
Lower body strength	Back squat, front squat, goblet squat, Romanian deadlift, conventional deadlift, reverse lunge, lateral lunge, hip hinge
Upper body strength	Push-up, diamond push-up, dip, overhead press form, lateral raise, bicep curl, tricep extension
Physiotherapy rehab	Shoulder external rotation, knee extension, hip abduction, clam shell, glute bridge, bird-dog, cat-cow, wall slide
Core & stability	Plank hold, side plank, dead bug, pallof press, hollow hold, Copenhagen plank
Yoga & mobility	Warrior I, warrior II, tree pose, downward dog, pigeon pose, chair pose, mountain pose

8. AI Coaching Feedback (Gemini 1.5 Flash)

8.1 When Feedback Is Generated

KineticAI generates **two types of AI feedback** using the Gemini 1.5 Flash free API (1 million tokens/day free):

- Real-time micro-cues: simple rule-based alerts (< 5ms, no API call) shown as text overlays during the set — e.g. 'Knees caving inward — push them out'
- Post-set AI coaching: after each set completes, session data (per-rep form scores, joint angles, fatigue indicators) sent to Gemini for a natural language coaching paragraph
- Session summary: at end of workout, full session stats sent to Gemini for a comprehensive coaching report with prioritised correction plan

8.2 Gemini Prompt Architecture

```
// Post-set coaching prompt
const prompt = `
You are an expert movement coach and physiotherapist.
Analyse this exercise set data and provide 2-3 specific, actionable
coaching tips.
Be encouraging but precise. Focus on the most impactful corrections only.

Exercise: ${exercise.name}
Reps completed: ${session.repCount}
Average form score: ${session.avgFormScore}/100
Primary issue: ${session.topIssue}  (occurred in ${session.issueFrequency}%
of reps)
```

```
Joint angle data:
- ${exercise.primaryJoint}: avg ${session.avgPrimaryAngle}° (target: $
{exercise.goldenRange}°)
- Symmetry: ${session.symmetryScore}% (left vs right deviation)
- Depth achieved: ${session.depthAchieved ? 'Yes' : 'No — ' +
session.avgDepthShortfall + '° short'}
- Fatigue detected: ${session.fatigueDetected ? 'Yes — form degraded from
rep ' + session.fatigueOnset : 'No'}

Response format: 2-3 bullet points, max 30 words each. Be specific to the
numbers above.
`;
```

8.3 Feedback Cost Management

Gemini 1.5 Flash is free for **1 million tokens/day**. A typical KineticAI session (5 sets × 3 API calls per set) uses approximately 1,500–2,500 tokens. The daily free quota handles **400–666 full workout sessions per day** — far exceeding any personal project load.

> 💡 **Graceful Fallback**
> If the Gemini API is unavailable or the user is offline, KineticAI falls back to rule-based feedback: a lookup table of pre-written coaching cues mapped to the specific issue detected (depth, symmetry, speed, alignment). The AI feedback enhances the experience but is never a dependency for core functionality.

9. Session Report — PDF Generation

9.1 Report Architecture

Session reports are generated **entirely client-side** using **jsPDF** — no server round-trip, no data uploaded. The user can download the PDF or share it via the Web Share API (on mobile) or copy a URL to a locally-stored blob. Reports are suitable for sharing with a physiotherapist or personal trainer.

9.2 Report Contents

Section	Contents	Data Source
Session header	Date, exercise, duration, device, model used	Session metadata
Summary metrics	Total reps, avg form score, best rep score, sets completed, calories est.	Computed from rep log

Per-rep timeline	Line chart: form score over all reps. Trend line shows fatigue	Rep log array
Joint angle breakdown	Table: per-joint avg angle, target range, deviation, pass/fail	JointAngleEngine output
Worst rep analysis	Screenshots of lowest-scoring rep with annotated skeleton overlay	Canvas frame capture
Best rep analysis	Screenshot of highest-scoring rep — visual reference for the user	Canvas frame capture
AI coaching summary	Gemini-generated natural language coaching report for the session	Gemini API response
Exercise prescription	Next session recommendation: reps, sets, focus areas	Rule-based engine + Gemini
Progress history (optional)	If localStorage has prior sessions: form score trend across sessions	localStorage

```
// Client-side PDF generation — jsPDF
import jsPDF from 'jspdf';

function generateSessionReport(session) {
  const pdf = new jsPDF({ orientation: 'portrait', unit: 'mm', format: 'a4'
});

  // Header
  pdf.setFont('helvetica', 'bold');
  pdf.setFontSize(24);
  pdf.text('KineticAI — Session Report', 20, 25);

  // Metrics section
  pdf.setFontSize(12);
  pdf.text(`Exercise: ${session.exerciseName}`, 20, 45);
  pdf.text(`Reps: ${session.totalReps}   Avg Form: $
{session.avgFormScore}/100`, 20, 52);

  // Embed best-rep canvas frame
  const imgData = bestRepCanvas.toDataURL('image/jpeg', 0.85);
  pdf.addImage(imgData, 'JPEG', 20, 60, 80, 60);

  // AI coaching section
  pdf.text('Coaching Feedback:', 20, 130);
  pdf.setFont('helvetica', 'normal');
  const lines = pdf.splitTextToSize(session.aiCoaching, 170);
  pdf.text(lines, 20, 137);

  pdf.save(`kineticai-${session.exerciseName}-${session.date}.pdf`);
}
```

10. Complete Free Tech Stack

💰 **Total Monthly Cost: $0.00**

KineticAI is a pure frontend application on Vercel. There is no backend, no database, no paid API required for core functionality. Gemini 1.5 Flash (1M tokens/day free) is the only external API used, and all fallback logic means the app works fully offline too.

10.1 Inference & ML

Tool	Free Tier / License	Role in KineticAI
TensorFlow.js	Open source (Apache 2.0)	Core ML runtime: loads MoveNet in browser, manages WebGPU/WebGL/WASM backends
MoveNet Lightning / Thunder	Free, Google, TF Hub	17-keypoint pose model: 50–120 FPS on WebGPU, 30–50 FPS on WebGL
MediaPipe BlazePose (TFJS)	Free, Google, open source	33-keypoint 3D-aware model for precision physiotherapy exercises
WebGPU API	Built into modern browsers, free	GPU acceleration for TF.js: Chrome 113+, Edge, Safari 18+, Firefox 119+
ONNX Runtime Web (optional)	Open source, Microsoft	Alternative inference runtime for custom fine-tuned models if added later

10.2 AI Coaching & Feedback

Tool	Free Tier	Role
Gemini 1.5 Flash API	1,000,000 tokens/day free	Post-set coaching paragraphs, session summaries, exercise prescriptions
Rule-based coaching engine	Built-in JavaScript, free	Real-time micro-cues (< 5ms), offline fallback coaching when Gemini unavailable

10.3 Frontend, Rendering & Output

Tool	Free Tier / License	Role

HTML5 Canvas API	Browser built-in, free	Real-time skeleton overlay rendering at 60 FPS (separate from inference loop)
WebRTC (getUserMedia)	Browser built-in, free	Camera access — works on desktop, laptop, phone with any browser
jsPDF	Open source (MIT)	Client-side PDF generation for session reports — no server needed
Chart.js	Open source (MIT)	Form score trend charts inside the PDF and dashboard view
Tailwind CSS (CDN)	Free CDN	Responsive UI styling — works without a build step
Vercel	Free hobby tier	Static frontend hosting, zero config CI/CD from GitHub, custom domain

11. Detailed Data Flow

11.1 Real-Time Inference Loop

1. **Step 1:** User opens KineticAI in browser. TF.js loads. WebGPU backend detected and initialised.
2. **Step 2:** User selects exercise from library. Exercise config JSON loaded (model, joints, golden ranges, cues).
3. **Step 3:** Model auto-selected (MoveNet Lightning or BlazePose). Model fetched from TF Hub CDN (~2–4MB). Cached in browser after first load.
4. **Step 4:** Camera permission granted. MediaStream initialised. Video element begins receiving frames.
5. **Step 5:** requestAnimationFrame loop starts: video frame → TF.js input tensor → MoveNet/BlazePose inference → 17/33 keypoints with confidence scores.
6. **Step 6:** JointAngleEngine receives keypoints. Calculates angles for all configured joint pairs using law-of-cosines.
7. **Step 7:** FormScorer compares current angles to golden ranges. Computes per-joint deviation. Updates rolling form score.
8. **Step 8:** RepCounter state machine receives primary joint angle. Updates phase (standing/descending/bottom/ascending). Fires rep event when complete.
9. **Step 9:** Canvas overlay renders skeleton (coloured joints: green = good, yellow = warning, red = critical), angle labels, rep count, form score bar.
10. **Step 10:** On rep completion: rep data appended to session log. Micro-cue displayed if any joint out of range.
11. **Step 11:** On set completion: session data packaged → Gemini API call → coaching paragraph rendered in side panel.

12. **Step 12:** On session end: full session log → PDF generation → download or share.

> ⏱️ **Latency Budget**
> Model inference (MoveNet Lightning, WebGPU): ~8ms. Joint angle calculation: < 1ms. Form scoring: < 1ms. Canvas render: ~2ms. Total analysis-to-visual-feedback loop: < 12ms at 30 FPS, < 6ms at 60 FPS. This is faster than the human eye can process (~40ms), meaning feedback is genuinely real-time with no perceptible lag. Gemini API call (post-set only): 1–3 seconds — this is async and non-blocking.

12. UI/UX Design

12.1 Application Layout

KineticAI uses a **split-screen layout** designed specifically for exercise use: the camera feed (with skeleton overlay) takes 70% of the screen, and a live coaching panel takes 30%. All controls are large enough to tap with a sweaty finger on a phone, and the critical metrics (rep count, form score) are rendered in large type visible from 2 metres away — the typical camera-to-user distance during exercise.

Screen 1 — Exercise Selection

- Grid of exercise cards with icons, category filters, and difficulty ratings
- Search bar for quick exercise lookup
- 'Custom mode' for adding any exercise with manual joint configuration
- Recent exercises section (localStorage) for quick repeat access

Screen 2 — Setup & Calibration

- Camera feed preview with framing guide overlay (dashed silhouette showing ideal camera distance/angle)
- Model selector: Lightning (fast) / Thunder (accurate) / BlazePose (3D)
- Set configuration: target reps, sets, rest time
- Auto-calibration: 5-second standing hold for baseline angle capture (personalises golden ranges to user's anatomy)

Screen 3 — Active Session

- Full-width camera feed with real-time skeleton overlay
- Top bar: rep counter (large), set counter, timer, FPS indicator, backend badge (WebGPU/WebGL/WASM)
- Joint angle labels rendered next to each joint (toggleable for cleaner view)

- Form score bar: colour gradient bar 0–100 updated every frame
- Side panel: latest coaching cue, phase indicator (DOWN / HOLD / UP), fatigue warning
- Per-rep log: small row of coloured dots (green/yellow/red) showing recent rep scores

Screen 4 — Session Review

- Rep-by-rep form score chart (line chart, Chart.js)
- Joint angle breakdown table
- AI coaching feedback paragraph (Gemini)
- Best rep and worst rep side-by-side frame comparison
- Download PDF button and Share button (Web Share API)

13. Accessibility & Privacy Architecture

13.1 Privacy-First by Design

🔒 **Zero Data Architecture**

KineticAI processes all video frames entirely on the user's device. No video, no keypoints, no joint angles, and no personal health data are ever transmitted to a server. The only outbound API call is the Gemini request — which contains only aggregated session statistics (numbers), never video, never images, never personally identifiable information. This means KineticAI is inherently GDPR-compliant, HIPAA-friendly, and appropriate for clinical use.

- No backend server — no data storage, no breach risk
- Gemini API receives: exercise name, rep count, angle averages — no video, no images, no names
- Session history stored in localStorage only — stays on the user's device
- PDF reports generated client-side — never uploaded
- Optional: local export of session JSON for personal record keeping

13.2 Accessibility Features

- High-contrast mode: joint colours, score bars, and overlays all meet WCAG 2.1 AA contrast ratios
- Screen reader announcements: rep completion and coaching cues announced via ARIA live regions
- Voice coaching mode: coaching cues spoken aloud via Web Speech API (free, browser built-in) so users can keep eyes on their form
- Large tap targets: all buttons minimum 44×44px for mobile use during exercise
- Reduced motion mode: skeleton animation and overlay transitions disabled per prefers-reduced-motion

14. Testing Strategy

14.1 Unit Tests

- Joint angle calculator: test calculateAngle() against known 3-point geometries (right angle, acute, obtuse, zero)
- Rep counter state machine: test all valid and invalid state transitions, edge cases (bounced rep, partial rep, over-range angle)
- Form scorer: test with mock keypoints at known angle deviations — verify weighted score output
- Fatigue detector: inject simulated angle degradation sequence and verify alert fires at correct threshold

14.2 Integration Tests

- Model load test: verify MoveNet Lightning and BlazePose both load and infer on a static test image, with correct keypoint counts
- End-to-end rep count test: play a pre-recorded squat video through the system, verify rep count matches human-labelled ground truth
- Gemini prompt test: send mock session data, verify response contains coaching language and is non-empty
- PDF generation test: complete a mock session, generate PDF, verify all required sections are present

14.3 Performance Tests

- FPS benchmark: measure actual inference FPS on Chrome/Edge/Safari across three device tiers (MacBook, mid-range laptop, budget Android phone)
- Memory leak test: run 30-minute session, monitor JS heap growth, verify no unbounded growth
- Cold start test: measure time-to-first-inference after page load (target: < 3 seconds for MoveNet Lightning)

14.4 Accuracy Evaluation

Evaluation	Benchmark
Keypoint accuracy vs. ground truth	Test on COCO keypoint validation set: compare MoveNet output to labelled keypoints
Rep count accuracy	Record 50 reps of 5 exercises (10 reps each), manually count, compare to KineticAI count. Target: < 5% error

Form score correlation	Have 5 volunteers perform sets at 3 quality levels (good/medium/poor as judged by a trainer). Verify KineticAI scores correlate (r > 0.8)
Angle accuracy	Use goniometer measurements on video as ground truth, compare to KineticAI angles. Target: < 5° RMS error

15. Milestone Plan (9-Week Roadmap)

Week	Milestone	Key Deliverables
Week 1	Foundation	Vercel project setup, TF.js + WebGPU backend init, camera feed, MoveNet Lightning inference loop, skeleton overlay on canvas, FPS counter
Week 2	Joint Angle Engine	calculateAngle() function, all 12 tracked joint pairs implemented, angle labels rendered on skeleton overlay, confidence filtering
Week 3	Exercise Library v1	JSON config schema, 10 exercises configured (5 strength, 3 physio, 2 yoga), exercise selection UI, golden range configuration
Week 4	Rep Counter	State machine for all 10 exercises, rep event system, rep counter display, per-rep form score, rep log array
Week 5	Form Scorer + Fatigue	5-component form score formula, per-joint deviation scores, fatigue detector (depth/symmetry/speed), coaching micro-cues overlay
Week 6	BlazePose Mode	BlazePose model integration, 3D keypoint handling, auto-model selector per exercise, mode toggle UI
Week 7	Gemini Coaching + Calibration	Post-set Gemini API integration, prompt engineering, offline fallback cues, auto-calibration (5s standing baseline)
Week 8	PDF Session Report	jsPDF integration, canvas frame capture, Chart.js score chart, all report sections, download + Web Share API
Week 9	Polish + Launch	Remaining 10 exercises,

		accessibility (voice coaching, ARIA), mobile optimisation, README + demo video, Vercel deploy + launch

16. Project Structure & Setup

16.1 Directory Structure

```
kineticai/
├── public/
│   ├── index.html              # Single-page app entry
│   └── assets/                 # Icons, exercise thumbnails
├── src/
│   ├── main.js                 # App bootstrap, backend detection
│   ├── inference/
│   │   ├── modelLoader.js      # TF.js model loading (MoveNet + BlazePose)
│   │   ├── poseDetector.js     # requestAnimationFrame inference loop
│   │   └── backendSelector.js  # WebGPU/WebGL/WASM fallback logic
│   ├── analysis/
│   │   ├── jointAngleEngine.js # calculateAngle(), all joint pair
computations
│   │   ├── repCounter.js       # State machine per exercise
│   │   ├── formScorer.js       # Weighted 5-component form score
│   │   ├── fatigueDetector.js  # Rep-over-rep degradation analysis
│   │   └── calibration.js      # 5-second standing baseline capture
│   ├── exercises/
│   │   ├── schema.js           # Exercise config JSON schema + validation
│   │   └── library/            # One JSON file per exercise (20+ files)
│   │       ├── squat.json
│   │       ├── pushup.json
│   │       └── ...             # All 20+ exercises
│   ├── rendering/
│   │   ├── skeletonRenderer.js # Canvas overlay: skeleton + angles +
colours
│   │   └── overlayHUD.js       # Rep count, form score bar, cue text
│   ├── coaching/
│   │   ├── geminiCoach.js      # Gemini API integration + prompt builder
│   │   └── ruleBasedCues.js    # Offline fallback coaching lookup table
│   ├── report/
│   │   ├── pdfGenerator.js     # jsPDF session report builder
│   │   └── chartBuilder.js     # Chart.js rep score chart for PDF
│   └── ui/
│       ├── exerciseSelector.js # Exercise grid + filter + search
│       ├── activeSession.js    # Session screen controller
│       └── sessionReview.js    # Post-session review screen
```

```
├── tests/
│   ├── unit/                    # Jest unit tests
│   └── integration/             # Playwright browser tests
├── vercel.json                  # Vercel deployment config
└── package.json
```

16.2 Local Setup

```
git clone https://github.com/ninjacode911/kineticai
cd kineticai
npm install

# Set Gemini API key (free — get at ai.google.dev)
echo 'VITE_GEMINI_API_KEY=AIza...' > .env.local

# Start dev server
npm run dev
# App opens at http://localhost:5173
# Allow camera permission when prompted

# Deploy to Vercel
vercel deploy
```

17. Risks & Mitigations

Risk	Likelihood	Mitigation
WebGPU not supported on target device	Medium	Automatic fallback: WebGL (30–50 FPS) → WASM (15–25 FPS). Status bar shows current backend. Core functionality works on all backends.
MoveNet keypoint confidence too low in poor lighting	High	Confidence threshold filter: keypoints below 0.3 confidence shown in grey and excluded from angle calculation. On-screen lighting guidance shown.
Camera permission denied by user	Low	Permission denied state shown with clear instructions. Option to upload a recorded video file instead of live camera.
Gemini API rate limit exceeded	Very Low	1M tokens/day free = 400+ full sessions. But: rule-based fallback always available. No core feature depends on Gemini.

Inaccurate angles due to camera angle	High	Calibration step + on-screen framing guide (dashed silhouette) ensures correct camera placement. Angle is always 2D projection — document this limitation clearly.
jsPDF PDF generation fails on some browsers	Low	Test on Chrome, Firefox, Safari. Fallback: export session data as JSON. Offer screenshot capture of report as secondary option.
User applies medical use-case incorrectly	Medium	Prominent disclaimer: 'KineticAI is a fitness coaching tool, not a medical device. Consult a physiotherapist before beginning a rehabilitation programme.' Displayed on first use and in every PDF.

18. Success Metrics

18.1 Technical Metrics

Metric	Target
Inference FPS on modern laptop (WebGPU, MoveNet Lightning)	≥ 50 FPS
Inference FPS on mid-range Android phone (WebGL)	≥ 25 FPS
Cold start time (page load to first inference)	< 3 seconds
Rep count accuracy on test video dataset	< 5% error rate
Form score correlation with expert assessment	Pearson r > 0.75
PDF generation time	< 2 seconds client-side

18.2 Portfolio & Career Metrics

Metric	Target
GitHub stars	≥ 200 within 4 weeks of launch
Vercel unique visitors (first month)	≥ 2,000

Demo video views (YouTube/LinkedIn)	≥ 5,000 in first 2 weeks
LinkedIn post impressions	≥ 10,000
HackerNews Show HN	Top 10 ranking within 24 hours

19. Future Roadmap (Post-Launch)

Phase 2 — Video Upload Analysis (Weeks 10–12)

- Allow users to upload a pre-recorded workout video instead of live camera
- Process at 2× speed (offline, not real-time): more thorough analysis, frame-by-frame worst-rep identification
- Particularly useful for coaches reviewing athlete video asynchronously

Phase 3 — Progress Tracking (Weeks 13–16)

- IndexedDB session history: store all sessions locally (no server needed)
- Progress dashboard: form score trend, range of motion improvement, rep count over time per exercise
- Streak tracking and workout calendar — gamification for adherence

Phase 4 — Custom Exercise Builder

- UI for defining a custom exercise: select joints to track, set golden ranges, set coaching cues
- Export custom exercise as JSON — shareable between users
- Physiotherapist mode: a physiotherapist can pre-configure an exercise library for their patient and share a URL

Phase 5 — Multi-Person & Group Classes

- MoveNet Multipose Lightning: track up to 6 people simultaneously
- Group class mode: show all users' form scores on a single display
- PT remote session: physiotherapist watches patient's live feed (WebRTC peer connection) with form overlay

20. Why KineticAI Stands Out in 2026

🏆 **The 2026 Hiring Signal**

KineticAI demonstrates a rare portfolio triple: (1) WebGPU and browser-native ML inference — the cutting-edge frontier of client-side AI that almost no engineer has shipped; (2) a clinical-grade application in a $134B market, validated by 2026 peer-reviewed research; (3) a complete, polished, publicly accessible product at a URL that any interviewer can open and immediately understand. The demo speaks for itself — within 10 seconds of opening, a recruiter can see their own skeleton being tracked at 60 FPS. That is a memorably impressive interview artifact.

Summary of What This Project Proves

- WebGPU mastery: you built and shipped a WebGPU-accelerated ML inference pipeline — less than 0.1% of engineers have done this
- Production browser AI: TF.js, model loading, backend selection, requestAnimationFrame loop — the full stack of real-time browser ML
- Domain knowledge: understanding joint angles, rep counting state machines, fatigue detection shows cross-domain thinking (ML + biomechanics)
- Clinical awareness: 2026 research-backed design decisions (BlazePose for 3D, calibration step, symmetry tracking) show you read the literature
- Privacy-by-architecture: zero-server design for sensitive health data demonstrates engineering maturity and real-world responsibility
- Full-stack product thinking: from inference engine to PDF report to accessibility to mobile UX — you shipped the whole product, not just the ML
- Connected to your prior work: YOLOv11 + WebGPU hemorrhoid detector → KineticAI tells a coherent story of deep WebGPU expertise across two shipped products

KineticAI · Navnit Amrutharaj · navnita004@gmail.com · github.com/ninjacode911